23 October 2026

Editor-in-Chief

Expert Systems with Applications

Dear Editor-in-Chief,

We submit the manuscript "Historical review under finite monitoring retention: Complete compression, semantic summaries and service boundaries" for consideration as a research article in Expert Systems with Applications.

Anomaly monitoring systems produce a detailed record of scores and threshold decisions. When an engineer revisits an alert, the question is practical: when did the channel first cross its threshold, what value was logged, and can every high record in a later interval still be retrieved? Retention and compression decide whether those answers survive. Prior work has optimized detection accuracy, alarm forecasting, compression ratio, or query latency. It leaves open a task level question: under one byte budget and one retention timeline, which retained representation can still answer which historical question? This gap matters because a summary that looks efficient can lose the first record or the complete report, and because a static ranking can reverse once older history expires.

The manuscript formalizes two executable review tasks: recovery of the first high record with its original numerical value, and recovery of every high record in a requested interval with a completeness flag. It compares six retention designs under one 16 byte record contract and identical byte budgets. The empirical base contains 116 histories from 28 server devices and two ESA missions, 1,392 serialized archives, 32 chronological conditions, and 5,280 online queries. Every format is charged for its schema, provenance, usability map, index metadata, and numerical payload. The online experiment applies the same arrivals, capacity, and queries to complete storage, support preserving summaries, coverage summaries, and an adaptive policy. A separate exercise tests whether past byte growth can set the next quarter capacity.

Three results give the paper its practical value. Complete compression fits 57 of 60 coarse histories at 256 KiB and all 60 at 1 MiB, while 53 of 56 fine server histories fit 256 KiB. A Full if fits strategy reconstructs 2,247 first records and 2,174 complete reports from 2,329 positive queries, against 2,167 and 2,167 for complete storage alone. Under chronological retention, coverage summaries recover 56 of 134 first records on the constrained fine server, compared with 24 of 134 for an adaptive policy that favors recency. It also reports where past only forecasts produce false feasible and false infeasible capacity decisions. These are not isolated benchmark gains. They define when complete history is the simplest sufficient design, when a semantically complete summary extends the answerable service, and when the requested history is no longer available.

The distinctive contribution is the joint treatment of two different review outputs, actual serialized bytes for every representation, and chronological expiration on the same histories. It connects an established completeness standard to measured storage and retention costs and converts the results into a short decision procedure for system designers. It reports the cases where simple complete storage wins, where strong summaries help, and where an adaptive policy or a past only forecast fails. That combination gives readers a defensible basis for retention architecture.

We believe the manuscript fits the journal. Expert Systems with Applications publishes applied intelligent systems research that connects methods to deployment decisions. The paper evaluates a monitoring component that sits behind anomaly detection and alerting and addresses what evidence remains available after the detector has done its work. The results should interest readers working on anomaly detection, monitoring and observability, time series management, and decision support.

The manuscript is original, is not under consideration elsewhere, and all authors have approved it. The authors declare no competing interests. The replication materials include frozen service streams, execution scripts, serialized archives, per query results, and verification receipts. The manuscript includes the required disclosure of generative AI use.

Thank you for your consideration.

Sincerely,

The Authors